

# Belief Distortions, Asset Prices, and Unemployment Fluctuations

Do Lee

New York University

## Motivation: Unemployment Volatility Puzzle (Shimer 2005)

**Macro puzzle:** Standard search model generates low unemployment volatility

# Motivation: Unemployment Volatility Puzzle (Shimer 2005)

**Macro puzzle:** Standard search model generates low unemployment volatility

**Finance approach:** Hiring is a forward-looking investment with risky cash flows

- ▶ Firms post vacancies based on their valuation of a new hire
- ▶ Valuation depends on beliefs about:
  - Expected cash flows: Future profits generated by the worker
  - Discount rates: Risk-adjusted present value of future profits

# Motivation: Unemployment Volatility Puzzle (Shimer 2005)

**Macro puzzle:** Standard search model generates low unemployment volatility

**Finance approach:** Hiring is a forward-looking investment with risky cash flows

- ▶ Firms post vacancies based on their valuation of a new hire
- ▶ Valuation depends on beliefs about:
  - Expected cash flows: Future profits generated by the worker
  - Discount rates: Risk-adjusted present value of future profits

**Standard model:** Limited sources of unemployment volatility

- ▶ Risk neutral  $\Rightarrow$  Constant discount rates
- ▶ Rational expectations  $\Rightarrow$  Ties beliefs to true cash flows, which are too stable

## **Subjective expectations:** This paper

- ▶ Subjective beliefs implied by survey forecasts overreact to cash flow news
  - Good news about cash flow leads to overoptimism  $\Rightarrow$  Inflates value of new hire
- ▶ Explains both aggregate volatility and dispersion across firms
  - Overreaction applies to both aggregate and idiosyncratic news

## **Rational expectations:** Hall 2017, Borovičková-Borovička 2017, Kehoe et al. 2023, ...

- ▶ Time-varying discount rates under full-information rational expectations
  - Recessions bring high discount rates  $\Rightarrow$  Depresses value of new hire
- ▶ Cross-sectional puzzle: In the data, firms respond strongly to idiosyncratic news
  - Hard to explain with rational discounting since idiosyncratic risk is largely diversifiable

**Question:** To what extent can belief overreaction explain labor market fluctuations?

**Approach:** Decompose variation in hiring using measured belief distortion

- ▶ Belief distortion  $\mathbb{F}_t - \mathbb{E}_t$ : Predictable mistakes in subjective beliefs
- ▶ Subjective belief  $\mathbb{F}_t$ : Survey forecast (equity research analysts)
- ▶ Objective belief  $\mathbb{E}_t$ : Machine learning forecast (neural network)

**Main findings:**

- ▶ Belief distortion measures survey overreaction (machine does not overreact)
- ▶ Belief distortion explains 90% of vacancy filling rate (68% of unemp.) variation
- ▶ Same mechanism drives firm-level hiring & return response to idiosyncratic news
- ▶ Search model with learning generates 60%+ of aggregate & firm-level variation

1. **Measure belief distortions:** Compare survey vs. machine learning forecasts
2. **Document overreaction:** Quantify survey overreaction using belief distortion
3. **Decompose labor market outcomes:** Time-series and cross-section
4. **Firm-level dynamics:** Estimate impulse response to idiosyncratic news
5. **Model:** Search and matching with learning about cash flows

## Labor market outcomes:

|              | Scope         | Measure                            |
|--------------|---------------|------------------------------------|
| Labor market | Time-series   | Vacancy filling rate (JOLTS)       |
|              | Cross-section | Firm-level hiring rate (Compustat) |

## Beliefs:

|                | Scope         | Subjective $\mathbb{F}_t$ | Objective $\mathbb{E}_t$ |
|----------------|---------------|---------------------------|--------------------------|
| Cash flows     | Time-series   | IBES analyst              | Machine learning         |
|                | Cross-section | IBES analyst              | Machine learning         |
| Discount rates | Time-series   | CFO forecast              | Machine learning         |
|                | Cross-section | IBES & Value Line         | Machine learning         |

**Time-series:**  $q_t$  = U.S. Vacancy filling rate for quarter  $t$  (Source: JOLTS, BLS)

$$q_t = \frac{f_t U_t}{V_t} = \frac{\text{Total Hires}}{\text{Total Job Vacancies}}$$

- ▶  $V_t$  job vacancies,  $U_t$  unemployment,  $f_t$  job finding rate (= total hires/unemployment)
- ▶ Countercyclical: High  $U_t$  relative to  $V_t$  during recession  $\Rightarrow$  High  $q_t$

**Cross-section:**  $hl_{i,t}$  = Hiring rate for firm  $i$  quarter  $t$  (Source: Compustat, JOLTS)

$$hl_{i,t} = \frac{L_{i,t} - (1 - \delta_{i,t-1})L_{i,t-1}}{L_{i,t-1}} = \frac{\text{Total Hires}}{\text{Total Employment}}$$

- ▶  $L_{i,t}$  employees at fiscal year-end, carried forward to quarterly
- ▶  $\delta_{i,t}$  job separation rate of firm  $i$ 's 2-digit NAICS industry
- ▶ **Firm-level sample:** Firms with common stocks (share codes 10, 11) on NYSE/AMEX/NASDAQ with IBES analyst coverage of expected earnings and stock price targets

**Cross-section:**  $E_{i,t}$  = Firm-level “street” earnings (IBES, 1983Q4–2023Q4):

- ▶ “Street”: Exclude one-off items not relevant to firm’s future operation
- ▶ Transformation to ensure positive values (Vuolteenaho 2002):

$$E_{i,t} = (1 - \lambda)E_{i,t}^* + \lambda r_t^f P_{i,t-1} > 0, \quad \lambda = 0.10$$

- Interpret as portfolio of 90% equity and 10% T-bills
- Allows  $\log(E_{i,t})$  to be well-defined when reported earnings negative  $E_{i,t}^* \leq 0$
- Apply similar transformation to firm-level stock returns

**Time-series:**  $E_t$  = Aggregate to S&P 500 level from firm-level

$$E_t = \Omega_t \sum_{i \in x_t} E_{i,t}^* / Divisor_t$$

- ▶  $x_t$  S&P 500 firms with IBES data,  $\Omega_t$  adjust for IBES coverage,  $Divisor_t$  S&P 500 divisor

**Cross-section:**  $\mathbb{F}_t[\Delta e_{i,t+h}]$  = IBES median consensus forecast of street earnings growth

- ▶ Respondents: Equity research analysts (1983Q4–2023Q4)
- ▶ Years  $h = 1, 2$ : Construct annual log growth forecast  $\mathbb{F}_t[\Delta e_{i,t+h}]$  from level forecast
  - Growth forecast:  $\mathbb{F}_t[\Delta e_{i,t+h}] \approx \log(\mathbb{F}_t[E_{i,t+h}]/\mathbb{F}_t[E_{i,t+h-1}])$
- ▶ Years  $h = 3, 4, 5$ : Interpret long-term growth (LTG) forecast as  $\mathbb{F}_t[\Delta e_{i,t+h}]$ 
  - LTG: Annualized growth forecast over next “three-to-five years”

**Time-series:**  $\mathbb{F}_t[\Delta e_{t+h}]$  = Aggregate to S&P 500 level from firm-level

- ▶ Years  $h = 1, 2$ : Aggregate using  $\mathbb{F}_t[E_{t+h}] = \Omega_t \sum_{i \in x_t} \mathbb{F}_t[E_{i,t+h}^*] \cdot S_{i,t} / \text{Divisor}_t$
- ▶ Years  $h = 3, 4, 5$ : Value-weighted average of LTG forecasts

**Time-series:**  $r_t$  = Log annual return on the S&P 500 with dividends

- Expected:  $\mathbb{F}_t[r_{t+h}]$  = CFO survey median consensus forecast (2001Q4–2023Q4)
  - Respondents: CFOs, VPs of finance, directors ( $\sim 1,600$  members as of 2022)
  - Horizon  $h$ : 1 and 10 years ahead; interpolate intermediate horizons linearly

**Cross-section:**  $r_{i,t}$  = Log annual return on firm  $i$ 's stock with dividends

- Expected:  $\mathbb{F}_t[r_{i,t+h}]$  from IBES & Value Line median consensus price target  $\mathbb{F}_t[P_{i,t+h}]$

$$\mathbb{F}_t[r_{i,t+h}] \approx \log \left( \frac{\mathbb{F}_t[P_{i,t+h}]}{P_{i,t}} + \frac{D_{i,t}}{P_{i,t}} \frac{\mathbb{F}_t[D_{i,t+h}]}{D_{i,t}} \right)$$

- Respondents: Equity research analysts (1999Q4–2023Q4)
- Horizon  $h$ : 1 year (IBES) and 5 years (Value Line), interpolate intermediate horizons
- Price  $P_{i,t}$  (CRSP), dividend  $D_{i,t}$  (Compustat),  $\mathbb{F}_t[D_{i,t+h}]/D_{i,t} \approx 1.064$  (postwar avg)

**Long Short-Term Memory (LSTM) neural network** (Bianchi, Lee, Ludvigson, Ma 2025):

$$\mathbb{E}_t[y_{t+h}] = G(\mathcal{X}_t, \beta_t^{TS}; \lambda_t^{TS}) \quad (\text{Time-Series})$$

$$\mathbb{E}_t[y_{i,t+h}] = G(\mathcal{X}_{i,t}, \beta_t^{CS}; \lambda_t^{CS}) \quad (\text{Cross-Section})$$

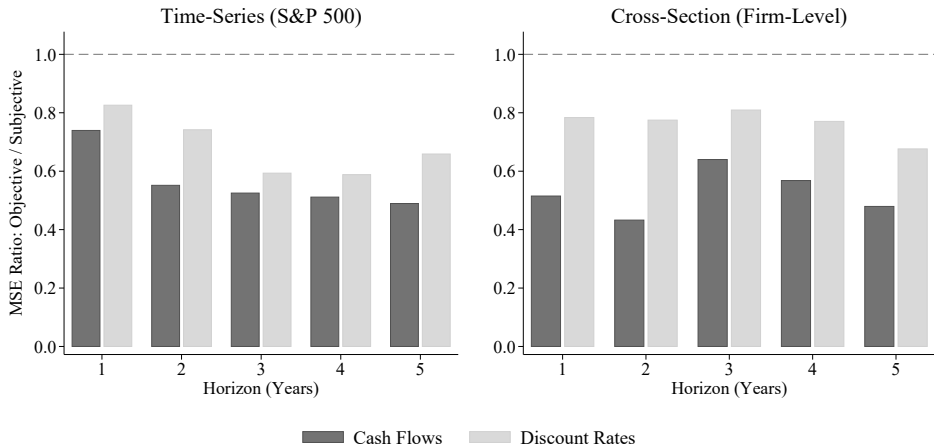
- Forecast target:  $y \in \{r, \Delta e\}$  at horizons  $h = 1, \dots, 5$  years
- Parameter  $\beta_t$ : Re-estimate quarterly (TS) or annually (CS) over rolling sample
- Regularization  $\lambda_t$ :  $L_1/L_2$  penalty, dropout, early stopping, ensemble average

**Input data:**

- Time-series:  $\mathcal{X}_t = \text{Real-time macro/financial, text (LDA from WSJ), survey data}$
- Cross-section:  $\mathcal{X}_{i,t} = [\mathcal{X}_t, \mathcal{C}_{i,t}]$  where  $\mathcal{C}_{i,t}$  includes firm characteristics (e.g. valuation, profitability, size, momentum, volatility), industry dummies (Chen-Zimmermann 2022)

# Machine Learning $\mathbb{E}_t$ More Accurate than Survey $\mathbb{F}_t$

- ▶ Machine MSE lower than survey MSE because survey makes predictable mistakes

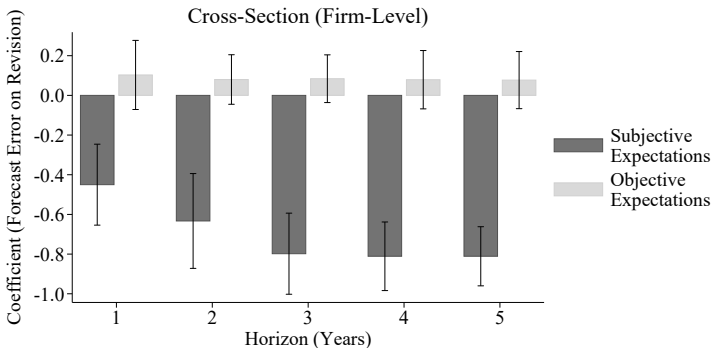


Notes: Figure shows relative forecast errors  $MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$  comparing machine learning to survey forecasts. Left panel: aggregate S&P 500 time-series forecasts. Right panel: cross-sectional forecasts across listed firms. Dark bars: cash flows (earnings growth); light bars: discount rates (stock returns). Out-of-sample testing period: 2005Q1–2023Q4.

# Survey $\mathbb{F}_t$ Overreacts, Machine Learning $\mathbb{E}_t$ Does Not

$$\overbrace{\Delta e_{i,t,t+h} - \mathbb{F}_t[\Delta e_{i,t,t+h}]}^{\text{Forecast Error}} = \beta \cdot \overbrace{[\mathbb{F}_t[\Delta e_{i,t,t+h}] - \mathbb{F}_{t-1}[\Delta e_{i,t,t+h}]]}^{\text{Forecast Revision}} + \overbrace{\alpha_i + \alpha_t}^{\text{Firm \& Time FE}} + \varepsilon_{i,t}$$

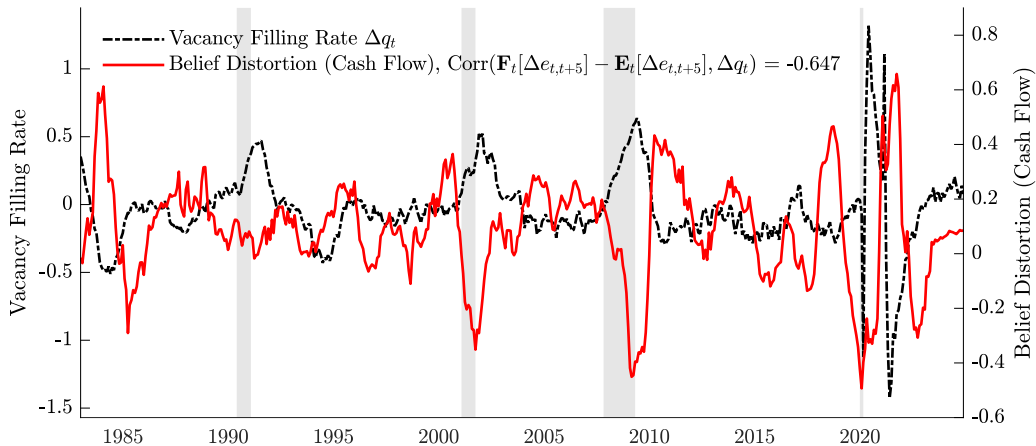
- ▶ Survey:
  - (+) forecast revision predicts (-) forecast error
  - ⇒ Overreaction
- ▶ Machine:
  - Forecast errors not predictable
  - ⇒ No overreaction



Notes: Figure reports  $\beta$  from firm-level Coibion and Gorodnichenko 2015 regressions of survey and machine forecast errors on forecast revisions with firm & time fixed effects. Sample: 2005Q1–2023Q4.

# Vacancy Filling Rate Tracks Belief Distortions in Cash Flows [► Details](#)

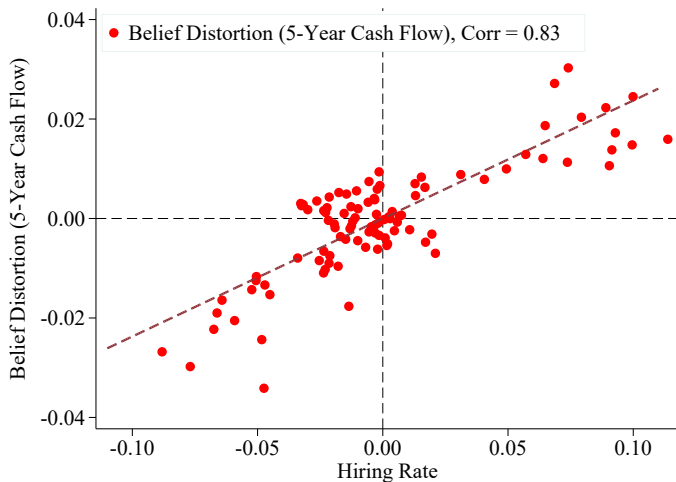
- Recession: Belief distortion  $\downarrow$  (pessimism), vacancy filling rate  $\uparrow$  (high  $U$  relative to  $V$ )



Notes: Left axis: Annual log change in the U.S. vacancy filling rate. Right axis: Belief distortion measured as expectational errors  $\mathbb{F}_t[\Delta e_{t,t+5}] - \mathbb{E}_t[\Delta e_{t,t+5}]$  in 5-year IBES survey forecasts of annualized S&P 500 earnings growth. Gray shaded areas indicate NBER recessions.

# Firms with Pessimistic Belief Distortions Hire Less [► Details](#)

- ▶ Cross-sectional dispersion after removing time & firm fixed effect
- ▶ Hiring rate =  
New hires / employment
- ▶ Belief distortion:  
Expectational errors  
 $\mathbb{F}_t[\Delta e_{i,t,t+5}] - \mathbb{E}_t[\Delta e_{i,t,t+5}]$



Notes: y-axis: belief distortion measured as expectational errors  $\mathbb{F}_t[\Delta e_{i,t,t+5}] - \mathbb{E}_t[\Delta e_{i,t,t+5}]$  in 5-year IBES survey forecasts of annualized S&P 500 earnings growth. x-axis: hiring rate. Each dot is a bin scatter representing one percentile across all observations in the sample. Sample: 2005Q1 to 2023Q4.

# Search and Matching Model (Diamond-Mortensen-Pissarides)

- ▶ Firm posts job vacancies  $V_t$  to attract unemployed workers  $U_t$ 
  - Matches formed at vacancy filling rate  $q_t$ , separated at rate  $\delta_t$
  - Posting a job vacancy costs  $\kappa$  per period
- ▶ Firm value  $\mathcal{V}$  satisfies Bellman equation:

$$\mathcal{V}(A_t, L_t) = \max_{V_t, L_{t+1}} \{E_t + \mathbb{F}_t[M_{t+1}\mathcal{V}(A_{t+1}, L_{t+1})]\}$$
$$s.t. \quad L_{t+1} = (1 - \delta_t)L_t + q_t V_t$$

- Cash flows (earnings):  $E_t = A_t L_t - W_t L_t - \kappa V_t$
- Productivity  $A_t$ , labor input  $L_t$ , wage rate  $W_t$
- Subjective expectation  $\mathbb{F}_t[\cdot]$ , stochastic discount factor  $M_{t+1}$

# Firm's Hiring Condition

- ▶ First-order condition: Firm equates cost of hiring with its expected discounted value

$$\frac{\kappa}{q_t} = \mathbb{E}_t \left[ M_{t+1} \frac{\mathcal{V}(A_{t+1}, L_{t+1})}{L_{t+1}} \right] = \frac{P_t}{L_{t+1}}$$

where  $P_t \equiv \mathbb{E}_t [M_{t+1} \mathcal{V}(A_{t+1}, L_{t+1})]$  ex-dividend market value (stock price)

- ▶ Take logarithms, rearrange terms, split the price-employment ratio  $\frac{P_t}{L_{t+1}}$ :

$$\log q_t = \log \kappa - \underbrace{\log \left( \frac{P_t}{E_t} \right)}_{\equiv pe_t} - \underbrace{\log \left( \frac{E_t}{L_{t+1}} \right)}_{\equiv el_t}$$

- Log-linearize price-earnings ratio  $pe_t \equiv \ln(P_t/E_t)$  around long-run mean  $\overline{pe}$

$$pe_t = c_{pe} - \mathbb{F}_t[r_{t+1}] + \mathbb{F}_t[\Delta e_{t+1}] + \rho \mathbb{F}_t[pe_{t+1}]$$

- $c_{pe}$  constant,  $\rho = \frac{\exp(\overline{pe})}{1+\exp(\overline{pe})} \approx 0.98$  time discount factor from log-linearization
  - $r_{t+1} = \log(\frac{P_{t+1}+E_{t+1}}{P_t})$  stock return, assuming firm pays out dividends = earnings
  - Same identity holds approximately if dividends  $\neq$  earnings e.g.,  $D_t \approx \frac{1}{2}E_t$
- Substitute recursively for the next  $h$  periods to obtain present value identity:

$$\underbrace{pe_t}_{\text{Price-Earnings}} = \underbrace{\sum_{j=1}^h \rho^{j-1} c_{pe}}_{\text{Constant}} - \underbrace{\sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[r_{t+j}]}_{\text{Discount Rate}} + \underbrace{\sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[\Delta e_{t+j}]}_{\text{Cash Flow}} + \underbrace{\rho^h \mathbb{F}_t[pe_{t+h}]}_{\text{Future Price-Earnings}}$$

# Decomposing the Vacancy Filling Rate

- Combine log-linearized  $pe_t$  with firm's hiring equation ( $c_q = \log \kappa - \sum_{j=1}^h \rho^{j-1} c_{pe}$ ):

$$\underbrace{\log q_t}_{\text{Vacancy Filling Rate}} = c_q + \underbrace{\sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[r_{t+j}]}_{\text{Discount Rate} \equiv \mathbb{F}_t[r_{t,t+h}]} - \underbrace{\left[ el_t + \sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[\Delta e_{t+j}] \right]}_{\text{Cash Flow} \equiv \mathbb{F}_t[e_{t,t+h}]} - \underbrace{\rho^h \mathbb{F}_t[pe_{t+h}]}_{\text{Future Price-Earnings} \equiv \mathbb{F}_t[pe_{t,t+h}]}$$

- Vacancy filling rate  $q_t$  is high (recession) either because of:
- High discount rates (return required to justify hiring)
  - Low expected cash flows (profit from hiring)
  - Or low expected price-earnings (terminal value)

# Variance Decomposition: Subjective vs. Objective Expectations

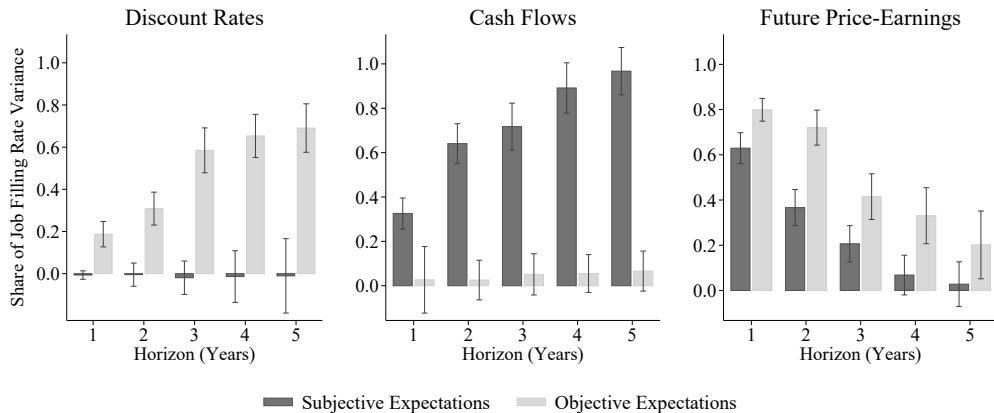
- Decomposition of  $Var [\log q_t]$  under subjective belief: Use survey forecast  $\mathbb{F}_t[\cdot]$

$$1 = \underbrace{\frac{Cov [\mathbb{F}_t[r_{t,t+h}], \log q_t]}{Var [\log q_t]}}_{\text{Discount Rate}} - \underbrace{\frac{Cov [\mathbb{F}_t[\Delta e_{t,t+h}], \log q_t]}{Var [\log q_t]}}_{\text{Cash Flow}} - \underbrace{\frac{Cov [\mathbb{F}_t[pe_{t,t+h}], \log q_t]}{Var [\log q_t]}}_{\text{Future Price-Earnings}}$$

- Estimate using OLS regression coefficients
  - Regress survey forecast  $\mathbb{F}_t[\cdot]$  on vacancy filling rate  $\log q_t$
  - Forecast horizons:  $h = 1, \dots, 5$  years
- Decomposition under objective belief:
  - Replace survey with machine forecast  $\mathbb{E}_t[\cdot]$
  - $\mathbb{F}_t - \mathbb{E}_t$  captures share that can be explained by belief distortion

# Variance Decomposition of the Vacancy Filling Rate

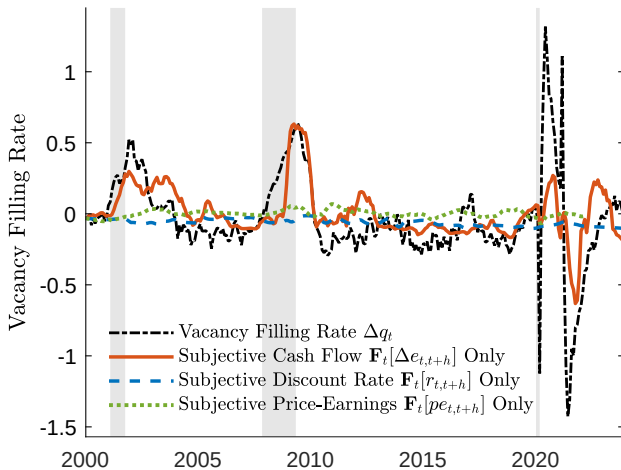
- Belief distortion  $\mathbb{F}_t - \mathbb{E}_t$  makes hiring sensitive to cash flow news



Notes: Sample: 2005Q1 to 2023Q4. Each bar shows Newey-West 95% confidence intervals with lags = 4 quarters.

# Role of Components in the Vacancy Filling Rate [► Details](#)

- ▶ Counterfactual series allowing one component to vary while holding others fixed
- ▶ Subjective cash flows play largest role in explaining vacancy filling rate variation



*Notes:* Counterfactual series are constructed by accumulating fitted values from regressions of vacancy filling rate growth on individual expectation measures at the  $h = 5$  year horizon, with all series initialized to the actual vacancy filling rate growth in 2001Q4.

# Why Do Some Firms Hire More Than Others?

- ▶ Define hiring rate at firm level:

$$hl_{i,t} \equiv \log \left( \frac{L_{i,t+1}}{L_{i,t}} - (1 - \delta_{i,t}) \right)$$

- $L_{i,t}$  employment (Compustat),  $\delta_{i,t}$  job separation rate from firm  $i$ 's industry (JOLTS)

- ▶ Decompose cross-sectional variance of hiring rate  $\text{Var}(\tilde{hl}_{i,t})$ :

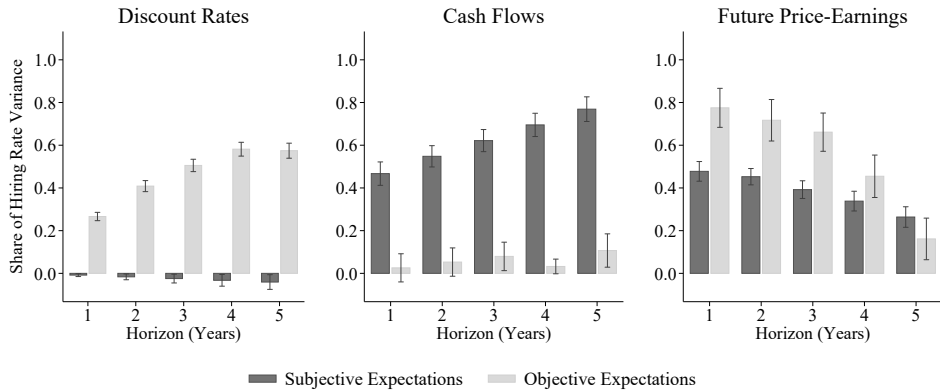
$$1 \approx - \underbrace{\frac{\text{Cov} \left( \mathbb{F}_t[\tilde{r}_{i,t,t+h}], \tilde{hl}_{i,t} \right)}{\text{Var}(\tilde{hl}_{i,t})}}_{\text{Discount Rate}} + \underbrace{\frac{\text{Cov} \left( \mathbb{F}_t[\tilde{e}_{i,t,t+h}], \tilde{hl}_{i,t} \right)}{\text{Var}(\tilde{hl}_{i,t})}}_{\text{Cash Flow}} + \underbrace{\frac{\text{Cov}(\mathbb{F}_t[\tilde{pe}_{i,t,t+h}], \tilde{hl}_{i,t})}{\text{Var}(\tilde{hl}_{i,t})}}_{\text{Future Price-Earnings}}$$

- $\tilde{x}_{i,t} = x_{i,t} - \sum_{i \in I} x_{i,t}$  cross-sectionally demeaned variable  $x_{i,t}$

- ▶ Estimate as coefficients from panel regressions with firm & time fixed effect

# Cross-Sectional Decomposition of Hiring Rate

- ▶ Belief distortion  $\mathbb{F}_t - \mathbb{E}_t$  makes hiring sensitive to cash flow news
  - Implies distortions can operate at firm level where actual hiring decisions are made



Notes: Sample: 2005Q1 to 2023Q4. Each bar shows 95% confidence intervals two-way clustered by firm and time.

# What is the Firm-Level Response to Idiosyncratic Shocks?

- ▶ Firm-level ( $i$ ) local projection of  $y_{i,t+h} = \log \text{employment}$   $\log(L_{i,t+h})$  and stock returns  $r_{i,t+h}$  on forecast revisions:

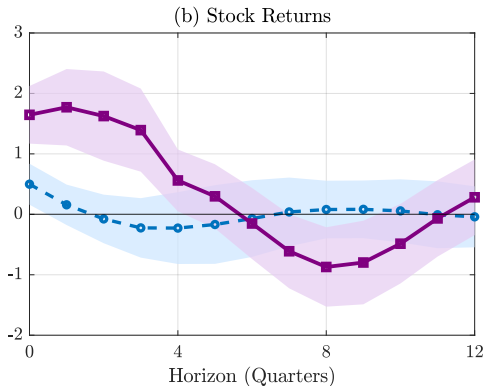
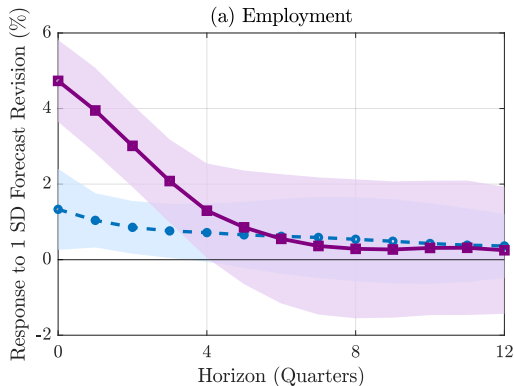
$$y_{i,t+h} = \beta_h \text{Revision}_{i,t} + \alpha_i + \alpha_{s(i),t} + \varepsilon_{i,t+h}, \quad s(i) \in \text{SIC2}$$

⇒ Dispersion in  $\text{Revision}_{i,t}$  captures belief response to idiosyncratic shock

- ▶ Subjective: Survey cash flow forecast revision  $\text{Revision}_{i,t} = \mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{F}_{t-1}[\Delta e_{i,t+1}]$   
⇒ Overreact to idiosyncratic shock, affect profits per worker by over- or under-hiring
- ▶ Objective: Machine cash flow forecast revision  $\text{Revision}_{i,t} = \mathbb{E}_t[\Delta e_{i,t+1}] - \mathbb{E}_{t-1}[\Delta e_{i,t+1}]$   
⇒ No overreaction, stable profits per worker

# Employment & Stock Returns Respond Only Under Subjective Beliefs

- After positive idiosyncratic shocks, firms with distorted subjective beliefs
  - (a) Adjust hiring excessively relative to firms with objective beliefs
  - (b) Experience stock returns that overshoot, then reverse as earnings disappoint



—●— Objective Expectations —■— Subjective Expectations

Notes: Blue (purple) line: IRF under objective (subjective) expectations proxied by machine (survey) forecasts. Shaded area: 90% confidence intervals two-way clustered by firm and time. Sample: 1984Q1–2023Q4.

# Model of Constant-Gain Learning: Environment

- ▶ True cash flow process has aggregate  $E_t$  and idiosyncratic  $\tilde{E}_{i,t}$  components

$$E_{i,t} = E_t \cdot \tilde{E}_{i,t} = \exp(e_t + \tilde{e}_{i,t})$$

- ▶ Log of each component follows an AR(1) process:

$$\begin{aligned} e_t &= \mu + \phi e_{t-1} + u_t, & u_t &\stackrel{iid}{\sim} \mathcal{N}(0, \sigma_u^2) \\ \tilde{e}_{i,t} &= \tilde{\mu}_i + \tilde{\phi} \tilde{e}_{i,t-1} + v_{i,t}, & v_{i,t} &\stackrel{iid}{\sim} \mathcal{N}(0, \sigma_v^2) \end{aligned}$$

- ▶ Log stochastic discount factor  $m_{t+1}$ :

$$m_{t+1} = -r_f - \frac{1}{2}\gamma^2\sigma_u^2 - \gamma u_{t+1}$$

with risk-free rate  $r_f$ , constant relative risk aversion  $\gamma$

# Model of Constant-Gain Learning: Subjective Beliefs

- ▶ Objectively, long-run mean identical across firms:  $\mu = \tilde{\mu}_i = 0$ 
  - But firm does not know true  $\mu$  and  $\tilde{\mu}_i$  in cash flow process
- ▶ Update beliefs using constant-gain learning rule: Learn at rate  $\nu$

$$\begin{aligned}\mathbb{F}_t[\mu] &= \mathbb{F}_{t-1}[\mu] + \nu (\Delta e_t - \mathbb{F}_{t-1}[\Delta e_t]) \\ \mathbb{F}_t[\tilde{\mu}_i] &= \mathbb{F}_{t-1}[\tilde{\mu}_i] + \nu (\Delta \tilde{e}_{i,t} - \mathbb{F}_{t-1}[\Delta \tilde{e}_{i,t}])\end{aligned}$$

- Start with true value  $\mathbb{F}_0[\mu] = \mathbb{F}_0[\tilde{\mu}_i] = 0$  so that  $\nu = 0$  case nests rational expectation
- ▶ Fading memory: Assigns smaller decaying weight on older observations
  - Generates overreaction as beliefs over-extrapolate from recent observations
  - Can arise from Bayesian updating under suspected regime switches (Adam, Marcet, and Nicolini 2016) or experience-based learning (Nagel and Xu 2021)

- Firm valuation: Equilibrium stock price  $P_{i,t}$  under subjective beliefs

$$P_{i,t} = \sum_{h=1}^{\infty} P_{i,t}^{(h)} = \sum_{h=1}^{\infty} \exp \left\{ A_i^{(h)} + B^{(h)} \mathbb{F}_t[\mu] + \tilde{B}^{(h)} \mathbb{F}_t[\tilde{\mu}_i] + \phi^h e_t + \tilde{\phi}^h \tilde{e}_{i,t} \right\}$$

- $P_{i,t}$  driven by belief distortions in cash flows  $\mathbb{F}_t[\mu]$ ,  $\mathbb{F}_t[\tilde{\mu}_i]$  (from fading memory)
- $P_{i,t}^{(h)}$  strip prices with recursively defined coefficients  $A_i^{(h)}$ ,  $B^{(h)}$ ,  $\tilde{B}^{(h)}$
- Hiring condition: Firms post vacancies until marginal cost equals marginal value

$$\underbrace{\frac{\kappa}{q_t}}_{\text{Cost of Hiring}} = \underbrace{\frac{P_{i,t}}{L_{i,t+1}}}_{\text{Value of New Hire}}$$

- Partial equilibrium: Exogenous cash flows absorb wage setting and worker belief
- Subjective valuation  $P_{i,t}$  drives both stock returns  $R_{i,t} = \frac{P_{i,t+1} + E_{i,t}}{P_{i,t}}$  and hiring  $L_{i,t+1}$

# Model Parameters

| Parameter      | Value | Description/Moments                                     |
|----------------|-------|---------------------------------------------------------|
| $\nu$          | 0.018 | Constant-gain learning (Malmendier and Nagel 2015)      |
| $\phi$         | 0.856 | Autocorrelation aggregate earnings growth               |
| $\sigma_u$     | 0.268 | S.D. aggregate earnings growth                          |
| $\tilde{\phi}$ | 0.698 | Autocorrelation firm-level earnings growth              |
| $\sigma_v$     | 0.194 | S.D. firm-level earnings growth                         |
| $r_f$          | 0.046 | Risk-free rate (De La O, Han, and Myers 2022)           |
| $\gamma$       | 1.586 | Average aggregate return (De La O, Han, and Myers 2022) |
| $\rho$         | 0.980 | Average price-earnings ratio                            |
| $B$            | 0.562 | Matching function efficiency (Kehoe et al. 2023)        |
| $\eta$         | 0.500 | Matching function elasticity (Kehoe et al. 2023)        |
| $\delta$       | 0.286 | Separation rate (Kehoe et al. 2023)                     |
| $\kappa$       | 0.314 | Per worker hiring cost (Elsby and Michaels 2013)        |

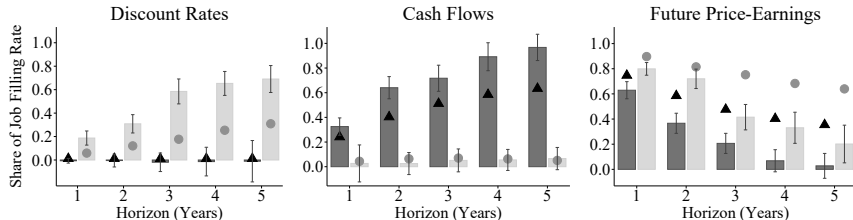
Notes: Table reports the parameter values used in the quantitative model along with the empirical moments they are calibrated to or sourced from. The model is calibrated at an annual frequency.

# Model vs. Data: Variance Decompositions

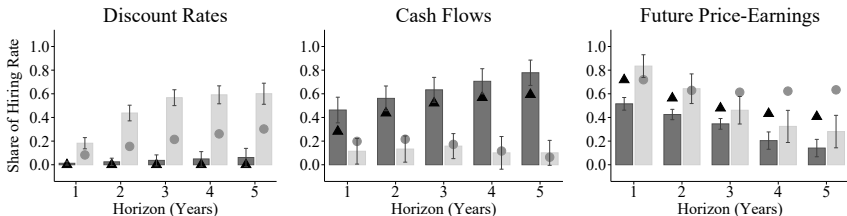
► Details (Labor Market)

► Model Parameters

(a) Time-Series  
Decomposition  
(Vacancy Filling Rate)



(b) Cross-Sectional  
Decomposition  
(Hiring Rate)



■ Subjective Expectations (Data)    ■ Rational Expectations (Data)  
▲ Subjective Expectations (Model)    ● Rational Expectations (Model)

Notes: Circle and triangle dots show the values of rational and subjective expectations implied by the model, respectively, derived from a simulation of 300 firms over 500 periods, with the first 150 periods discarded as a burn-in.

# Model vs. Data: Moments

- Learning improves model's ability to match asset market and labor market moments

(a) Asset Market

| Moment                                            | Data        | Learning Model | Rational Model |
|---------------------------------------------------|-------------|----------------|----------------|
| $SD(pe_t) \times 100$                             | <b>47.0</b> | <b>43.5</b>    | <b>13.0</b>    |
| $SD_i(pe_{i,t}) \times 100$                       | <b>22.6</b> | <b>21.1</b>    | <b>4.2</b>     |
| $SD(r_t) \times 100$                              | 16.0        | 12.3           | 3.0            |
| $SD(\mathbb{F}_t[r_{t+1}]) \times 100$            | 1.1         | 1.4            | 0.5            |
| $SD(\mathbb{F}_t[\Delta e_{t+1}]) \times 100$     | 26.8        | 24.3           | 7.2            |
| $SD_i(r_{i,t}) \times 100$                        | 5.7         | 3.1            | 1.2            |
| $SD_i(\mathbb{F}_t[r_{i,t+1}]) \times 100$        | 2.6         | 0.2            | 0.2            |
| $SD_i(\mathbb{F}_t[\Delta e_{i,t+1}]) \times 100$ | 14.0        | 16.6           | 3.9            |

(b) Labor Market

| Moment                      | Data         | Learning Model | Rational Model |
|-----------------------------|--------------|----------------|----------------|
| $SD(u_t) \times 100$        | <b>2.10</b>  | <b>1.28</b>    | <b>0.34</b>    |
| $SD_i(hl_{i,t}) \times 100$ | <b>15.70</b> | <b>10.39</b>   | <b>4.65</b>    |
| $SD(q_t) \times 100$        | 8.70         | 6.16           | 0.91           |
| $Corr(u_t, q_t)$            | -0.82        | -0.86          | -0.99          |

Notes:  $SD(\cdot)$  = time-series standard deviation.  $SD_i(\cdot)$  = cross-sectional standard deviation.  $pe_t$  = log price-earnings ratio,  $r_t$  = log stock return,  $\Delta e_t$  = log earnings growth,  $q_t$  = job-filling rate,  $u_t$  = unemployment rate,  $hl_{i,t}$  = firm-level hiring rate.  $\mathbb{F}_t[\cdot]$  = subjective expectations formed at time  $t$ . Data = empirical moments. Model (Learning) = constant-gain learning model. Model (Rational) = rational expectations benchmark.

- ▶ Subjective beliefs are distorted as they overreact to news
  - Comparing survey vs. machine learning forecasts uncovers these distortions
- ▶ Overreaction to cash flow news drives asset prices and hiring
  - Both in the time-series and the cross-section
  - Results consistent with learning about cash flow with fading memory
- ▶ Offers new perspective on unemployment volatility puzzle
  - Belief distortions drive value of new hire, driving unemployment fluctuations

# Appendix

- ▶ **Unemployment Volatility Puzzle:** Shimer 2005; Hagedorn and Manovskii 2008; Hall and Milgrom 2008; Pissarides 2009; Elsby and Michaels 2013; Kudlyak 2014; Chodorow-Reich and Karabarbounis 2016; Ljungqvist and Sargent 2017; Hall 2017; Borovičková and Borovička 2017; Kilic and Wachter 2018; Mitra and Xu 2019; Kehoe, Midrigan, and Pastorino 2019; Kehoe et al. 2023; Meeuwis et al. 2023
  - This paper: Reframes unemployment volatility as belief-driven
- ▶ **Labor Market Frictions and Asset Prices:** Merz and Yashiv 2007; Donangelo 2014; Belo, Lin, and Bazdresch 2014; Favilukis and Lin 2015; Kuehn, Simutin, and Wang 2017; Petrosky-Nadeau, Zhang, and Kuehn 2018; Donangelo et al. 2019; Liu 2021; Belo et al. 2023
  - This paper: Introduce subjective beliefs to explain differences in hiring across firms
- ▶ **Non-Rational Expectations and Business Cycles:** Marcet and Sargent 1989; Evans and Honkapohja 2001; Woodford 2001; Mankiw and Reis 2002; Sims 2003; Venkateswaran 2014; Coibion and Gorodnichenko 2015; Gabaix 2019; Ma et al. 2020; Acharya and Wee 2020; Bordalo et al. 2021; Bianchi, Ludvigson, and Ma 2022; Bianchi, Ilut, and Saijo 2023; Menzio 2023; Bhandari, Borovička, and Ho 2024; Fukui, Gormsen, and Huber 2024; Du et al. 2025; Bigio, Silva, and Zilberman 2025; Bloom, Codreanu, and Fletcher 2025
  - This paper: Clarify that bias in expected cash flow drives unemployment fluctuations
- ▶ **Non-Rational Expectations and Asset Prices:** Timmermann 1993; Barberis, Shleifer, and Vishny 1998; Chen, Da, and Zhao 2013; Greenwood and Shleifer 2014; Greenwood and Hanson 2014; Adam, Marcet, and Nicolini 2016; Giglio et al. 2021; De La O and Myers 2021; Nagel and Xu 2022; Barrero 2022; Jin and Sui 2022; De La O, Han, and Myers 2022; Binsbergen, Han, and Lopez-Lira 2022; Adam and Nagel 2023; Bianchi, Ludvigson, and Ma 2024; Bordalo et al. 2024; Décaire and Graham 2024; Chaudhry 2025
  - This paper: Show that biases drive both asset prices and real hiring decisions

Time-series: Vacancy filling rate  $q_t$  for quarter  $t$  (Source: JOLTS, BLS)

$$q_t = \frac{f_t U_t}{V_t}$$

- ▶  $V_t$  and  $U_t$ : U.S. job openings and unemployment level
- ▶  $f_t = 1 - \frac{U_t - U_t^s}{U_{t-1}}$ : Job finding rate, where  $U_t^s$  short-term unemployed

Cross-section: Hiring rate  $hl_{i,t}$  for firm  $i$  quarter  $t$  (Source: Compustat, JOLTS)

$$hl_{i,t} = \log \left( \frac{L_{i,t+1}}{L_{i,t}} - (1 - \delta_{i,t}) \right)$$

- ▶  $L_{i,t}$ : Annual employees, interpolate to quarterly using latest value
- ▶  $\delta_{i,t}$ : Job separation rate of firm  $i$ 's NAICS2 industry
- ▶ Sample: All common stocks (share codes 10, 11) on NYSE/AMEX/NASDAQ with IBES analyst coverage of earnings and stock price targets

- ▶ Realized return  $r_t$ : Annual log return on CRSP value-weighted index with dividend
- ▶ Expected return  $\mathbb{F}_t[r_{t+h}]$ :
  - Source: CFO survey median consensus forecast (2001Q4–2023Q4)
  - Respondents: CFOs, VPs of finance, directors (1,600 members as of 2022)
  - Prediction target: Aggregate S&P 500 stock returns
  - Horizon  $h$ : 1 and 10 years ahead; interpolate intermediate horizons linearly

- ▶ Firm-level earnings  $E_{i,t}^*$ : Street earnings for firm  $i$  (Source: IBES)
  - “Street”: Exclude discontinued operation, extraordinary charge, non-operating item
  - Construct from earnings per share:  $E_{i,t}^* = EPS_{i,t}^* \cdot S_{i,t}$  where  $S_{i,t}$  shares outstanding
  - Transformation to ensure positive earnings (Vuolteenaho 2002):

$$E_{i,t} = (1 - \lambda)E_{i,t}^* + \lambda r_t^f P_{i,t-1} > 0, \quad \lambda = 0.10$$

- ▶ Define firm as a portfolio of 90% equity & 10% 1-year T-bills (with rate  $r_t^f$ )
  - ▶ Allows  $\log(E_{i,t})$  to be well-defined when reported earnings negative  $E_{i,t}^* \leq 0$
- ▶ Firm-level expected log earnings growth  $\mathbb{F}_t[\Delta e_{i,t+h}]$ :
  - Source: IBES median consensus forecast (1983Q4–2023Q4)
  - Respondents: Equity research analysts

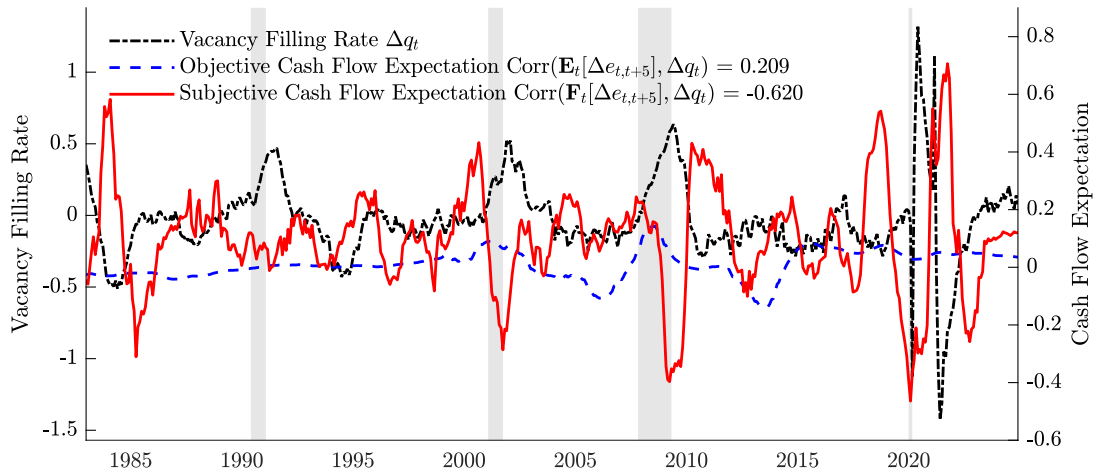
Long Short-Term Memory (LSTM) neural network (Bianchi et al. 2022, 2024, 2025)

$$\mathbb{E}_t[y_{t+h}] = G(\mathcal{X}_t, \beta_t; \lambda_t)$$

- ▶ Target:  $y_{t+h} \in \{r_{t+h}, \Delta e_{t+h}\}$  predicted  $h = 1, 2, \dots, 5$  years ahead
- ▶ Estimation:
  - Training: Rolling samples, parameters  $\beta_t$  re-estimated quarterly in real-time
  - Validation: Pseudo out-of-sample for hyperparameters  $\lambda_t$
  - Regularization:  $L_1/L_2$  penalties, dropout, early stopping, ensemble averaging
  - Architecture selection: Layers, neurons, training/validation window lengths
- ▶ Input data  $\mathcal{X}_t$ :
  - Macro/financial data, text (LDA factors from WSJ), macro data & FOMC surprises
  - Lagged survey forecast  $\mathbb{F}_{t-1}[y_{t+h-1}]$

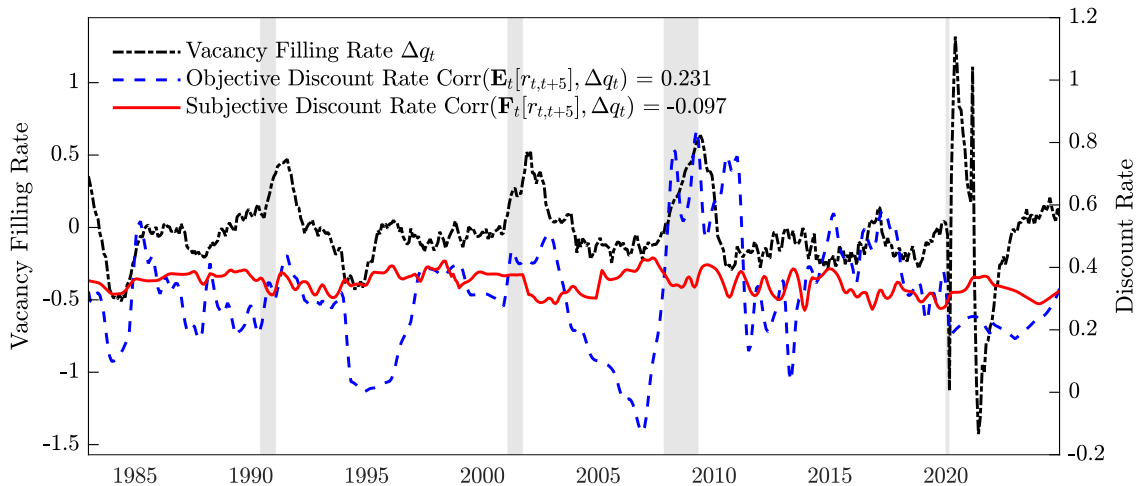
# Vacancy Filling Rate Tracks Belief Distortions in Cash Flows [▶ Return](#)

► Belief distortion: Expectational errors  $\mathbb{F}_t[\Delta e_{t,t+h}] - \mathbb{E}_t[\Delta e_{t,t+h}]$



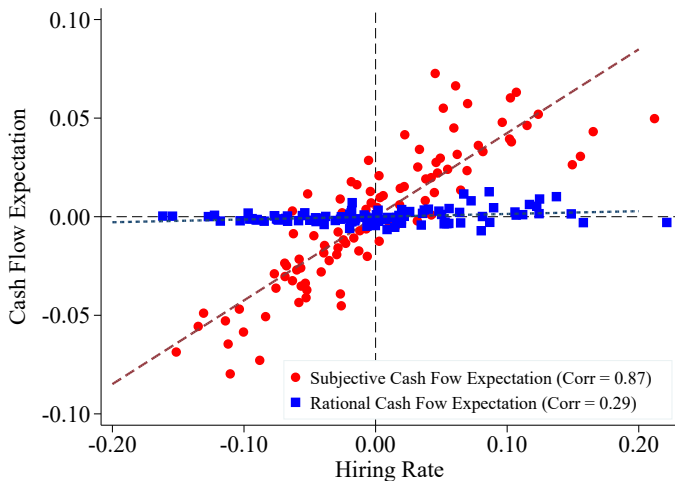
Notes: Left axis: Annual log change in the U.S. vacancy filling rate. Right axis: 5-year forecasts of annualized S&P 500 earnings growth. Subjective expectation: IBES median analyst projections for the next four fiscal years and long-term growth (LTG). Objective expectation: Machine learning forecasts from Long Short-Term Memory (LSTM) neural networks. Gray shaded areas indicate NBER recessions.

# Vacancy Filling Rate Tracks Objective Discount Rates

[Return](#)

# Firms with Pessimistic Belief Distortions Hire Less [▶ Return](#)

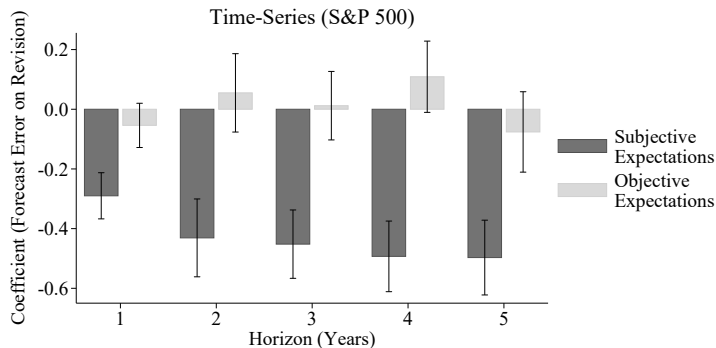
- ▶ Cross-sectional dispersion net of time & portfolio fixed effect
- ▶ Hiring rate =  
New hires / employment
- ▶ Belief distortion:  
Expectational errors  
 $\mathbb{F}_t[\Delta \tilde{e}_{i,t,t+h}] - \mathbb{E}_t[\Delta \tilde{e}_{i,t,t+h}]$



Notes: x-axis: Cross-sectionally demeaned real-time expectation of future earnings. Subjective Expectation:  $\mathbb{F}_t[\tilde{e}_{i,t,t+j}]$  based on IBES survey forecasts. Objective Expectation:  $\mathbb{E}_t[\tilde{e}_{i,t,t+j}]$  based on machine learning forecasts. y-axis: Cross-sectionally demeaned hiring rate. Each dot is a bin scatter representing one percentile across all observations in the sample. Sample: 2005Q1 to 2023Q4.

$$\overbrace{\Delta e_{t,t+h} - \mathbb{F}_t[\Delta e_{t,t+h}]}^{\text{Forecast Error}} = \beta \cdot \overbrace{[\mathbb{F}_t[\Delta e_{t,t+h}] - \mathbb{F}_{t-1}[\Delta e_{t,t+h}]]}^{\text{Forecast Revision}} + \varepsilon_t$$

- ▶ Survey:
  - (+) forecast revision predicts (-) forecast error
  - ⇒ Overreaction
- ▶ Machine:
  - Forecast errors not predictable
  - ⇒ No overreaction



Notes: Figure reports  $\beta$  from time-series Coibion and Gorodnichenko 2015 regressions of survey and machine forecast errors on forecast revisions for aggregate S&P 500 cash flows. Sample: 2005Q1–2023Q4.

- Log-linearize log price-dividend  $pd_t \equiv \ln(P_t/D_t)$  around long-term average  $\overline{pd}$

$$pd_t = c_{pd} + \Delta d_{t+1} - r_{t+1} + \rho pd_{t+1}$$

where  $c_{pd}$  constant,  $r_{t+1} \equiv \log\left(\frac{P_{t+1}+D_{t+1}}{P_t}\right)$  stock returns,  $\rho \equiv \frac{\exp(\overline{pd})}{1+\exp(\overline{pd})} = 0.98$

- Substitute in log price-earnings  $pe_t = pd_t + de_t$  where  $de_t$  log payout ratio

$$pe_t = c_{pd} + \Delta e_{t+1} - r_{t+1} + \rho pe_{t+1} + (1 - \rho)de_{t+1}$$

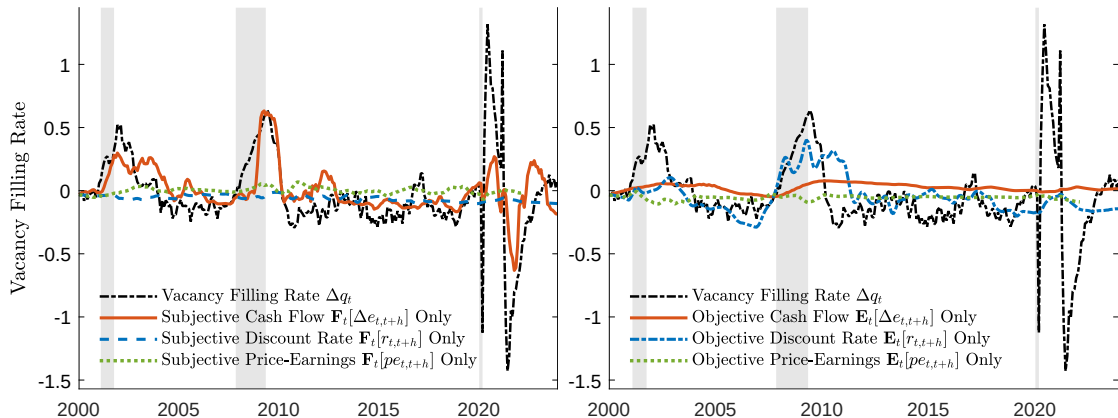
- Since  $1 - \rho \approx 0$  and  $de_t$  bounded, approximate  $(1 - \rho)de_{t+1}$  as a constant

$$pe_t \approx c_{pe} + \Delta e_{t+1} - r_{t+1} + \rho pe_{t+1}, \quad c_{pe} \approx c_{pd} + (1 - \rho)de_{t+1}$$

- Recursively substitute for the next  $h$  periods

$$pe_t = \sum_{j=1}^h \rho^{j-1} (c_{pe} + \Delta e_{t+j} - r_{t+j}) + \rho^h pe_{t+h}$$

# Role of Components in the Vacancy Filling Rate: $\mathbb{F}_t$ vs. $\mathbb{E}_t$ [Return](#)



Notes: Counterfactual series accumulate fitted values from regressions of vacancy-filling-rate growth on each expectation measure ( $h = 5$ ), initialized to the actual value in 2001Q4. Gray shaded areas indicate NBER recessions. Sample: 2000Q4–2023Q4.

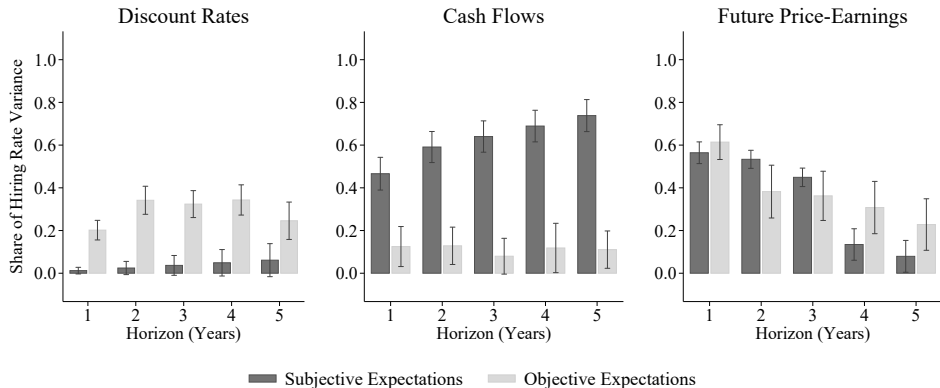
- ▶ Results are robust to using alternative surveys for earnings growth forecasts:
  - IBES, Bloomberg (BBG), CFO survey
  - Kalman-filtered (KF) composite of IBES, BBG, and CFO

| Horizon $h$ (Years)                                                                                                      | 1        | 2        | 3        | 4        | 5        |
|--------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|
| Subjective Expectations: $\log q_t = c_q + \mathbb{F}_t[r_{t,t+h}] - \mathbb{F}_t[e_{t,t+h}] - \mathbb{F}_t[pe_{t,t+h}]$ |          |          |          |          |          |
| (-) Cash Flow (KF)                                                                                                       | 0.578*** | 0.625*** | 0.684*** | 0.887*** | 0.933*** |
| $t$ -stat                                                                                                                | (3.046)  | (4.275)  | (4.894)  | (6.019)  | (7.612)  |
| (-) Cash Flow (BBG)                                                                                                      | 0.586*** | 0.830*** | 0.851*** | 0.896*** | 0.949*** |
| $t$ -stat                                                                                                                | (8.476)  | (8.317)  | (7.213)  | (5.288)  | (4.541)  |
| (-) Cash Flow (CFO)                                                                                                      | 0.637*   |          |          |          |          |
| $t$ -stat                                                                                                                | (1.934)  |          |          |          |          |

Notes: KF summarizes the alternative survey measures into a single series using a Kalman filter. The sample for BBG and KF is quarterly from 2006Q1 to 2023Q4. The sample for CFO is quarterly from 2005Q1 to 2019Q3. Newey-West  $t$ -statistics with lags = 4 in parentheses: \*sig. at 10%. \*\*sig. at 5%. \*\*\*sig. at 1%.

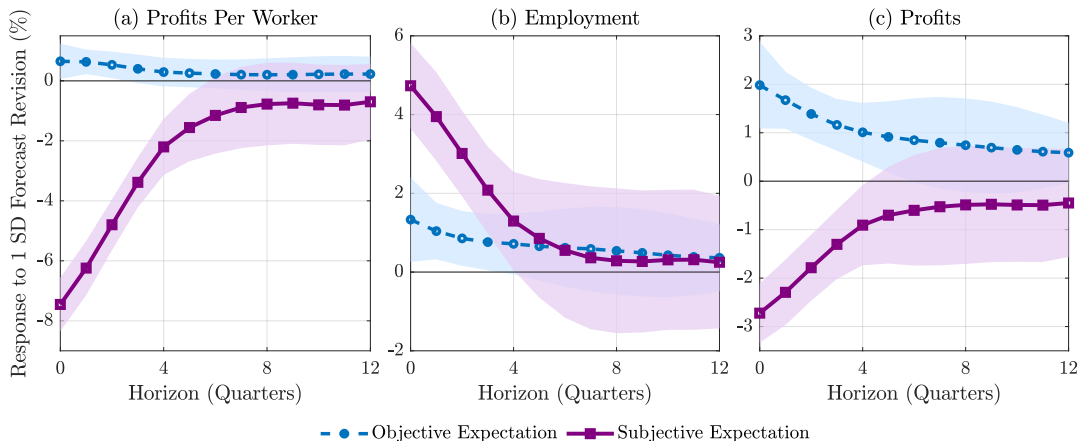
### ► Financial constraint proxies:

- Size, payout ratio, SA index (size and age), expected free cash flow (size, leverage, profitability, growth), Whited-Wu index (leverage, payout, size, growth)



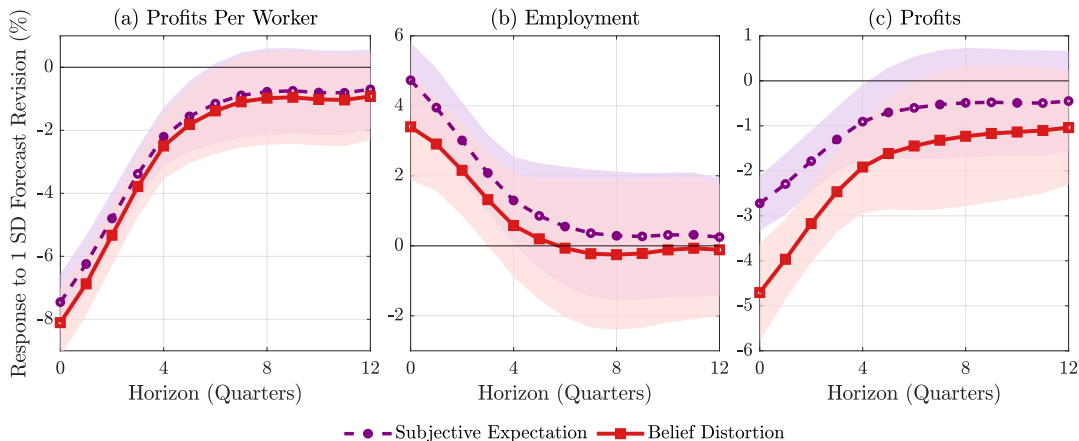
*Notes:* Figure estimates cross-sectional decomposition of hiring rate, controlling for measures of financial constraints. Financial constraint controls include firm size, payout ratio, SA index, expected free cash flow, and the Whited-Wu index. The sample is quarterly from 2005Q1 to 2023Q4. Each bar shows 95% confidence intervals clustered by portfolio and time.

# Employment Rise, Profits Fall Under Subjective Beliefs

[Return](#)

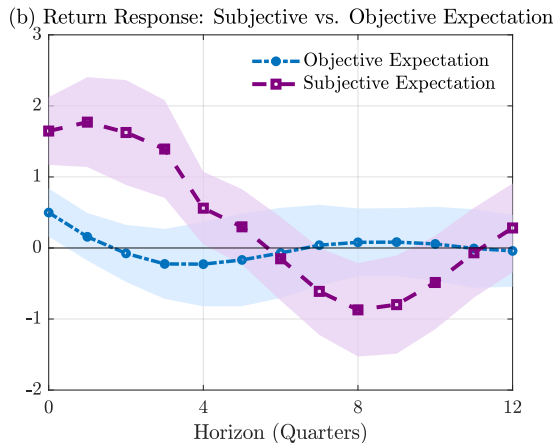
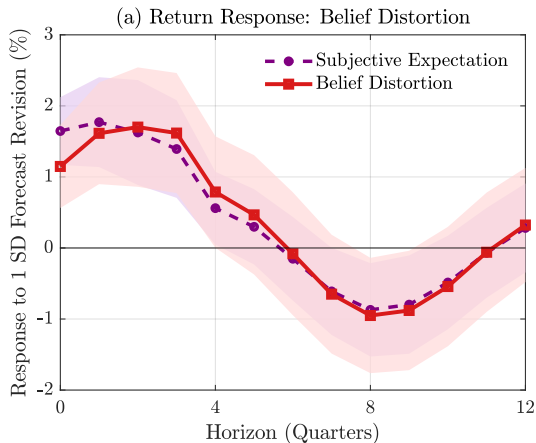
Notes: Blue (violet) line: IRF from revisions in objective (subjective) expectation. Shaded area: 90% confidence intervals two-way clustered by firm and time. Sample: 1984Q1–2023Q4.

# Subjective Belief Response Driven by Belief Distortion: Hiring

[Return](#)

Notes: Red (violet) line: IRF from revisions in belief distortion (subjective expectation). Shaded area: 90% confidence intervals two-way clustered by firm and time. Sample: 1984Q1–2023Q4.

# Subjective Belief Response Driven by Belief Distortion: Returns ► Return



Notes: Red (violet) line: IRF from revisions in belief distortion (subjective expectation). Shaded area: 90% confidence intervals two-way clustered by firm and time. Sample: 1984Q1–2023Q4.

- ▶ Aggregate strip price ( $h$ -period): Guess and verify recursive form

$$P_t^{(h)} = \mathbb{F}_t[M_{t+1}P_{t+1}^{(h-1)}] = \exp\{A^{(h)} + B^{(h)}\mathbb{F}_t[\mu] + \phi^h e_t\}$$

with coefficients:

$$A^{(h)} = A^{(h-1)} - r_f + \frac{1}{2}C^{(h)}[(1 + \nu^2)C^{(h)} - 2\gamma]\sigma_u^2$$

$$B^{(h)} = B^{(h-1)} + \phi^{h-1}, \quad C^{(h)} = \nu B^{(h-1)} + \phi^{h-1}, \quad A^{(0)} = B^{(0)} = C^{(0)} = 0$$

- ▶ Realized return on strip:

$$R_{t+1}^{(h)} = \frac{P_{t+1}^{(h-1)}}{P_t^{(h)}} = \exp\{[A^{(h-1)} - A^{(h)}] + C^{(h)}(\mu - \mathbb{F}_t[\mu] + u_{t+1})\}$$

- ▶ Subjective expected return on strip:

$$\mathbb{F}_t \left[ R_{t+1}^{(h)} \right] = \exp \{ r_f + C^{(h)} \gamma \sigma_u^2 \}$$

- Solve for  $q_t$  and  $L_{i,t+1}$  by iterating on labor market tightness  $\theta_t$  until convergence:

1. Initialize labor market tightness:  $\theta_t^{(0)} = 1$
2. Construct vacancy filling rate using Cobb-Douglas matching:  $q_t^{(s)} = B(\theta_t^{(s)})^{-\eta}$
3. Update each firm's employment policy using the hiring equation:

$$L_{i,t+1}^{(s)} = \frac{P_{i,t} q_t^{(s)}}{\kappa}$$

4. Update each firm's vacancy posting using the employment accumulation equation:

$$V_{i,t}^{(s)} = \frac{1}{q_t^{(s)}} (L_{i,t+1}^{(s)} - (1 - \delta) L_{i,t})$$

5. Aggregate firm-level variables over the set of firms  $I$ :

$$V_t^{(s)} = \sum_{i \in I} V_{i,t}^{(s)}, \quad L_{t+1}^{(s)} = \sum_{i \in I} L_{i,t+1}^{(s)}, \quad U_t^{(s)} = 1 - \sum_{i \in I} L_{i,t}$$

6. Update labor market tightness:  $\theta_t^{(s+1)} = \frac{V_t^{(s)}}{U_t^{(s)}}$ . Check convergence:  $|\theta_t^{(s+1)} - \theta_t^{(s)}| < \varepsilon$

# References I

- Acharya, Sushant and Shu Lin Wee (2020). "Rational Inattention in Hiring Decisions". In: *American Economic Journal: Macroeconomics* 12.1, 1–40.
- Adam, Klaus, Albert Marcet, and Juan Pablo Nicolini (2016). "Stock Market Volatility and Learning". In: *The Journal of Finance* 71.1, pp. 33–82. ISSN: 00221082, 15406261.
- Adam, Klaus and Stefan Nagel (2023). "Expectations data in asset pricing". In: *Handbook of Economic Expectations*. Ed. by Rüdiger Bachmann, Giorgio Topa, and Wilbert van der Klaauw. Academic Press, pp. 477–506. ISBN: 978-0-12-822927-9.
- Barberis, Nicholas, Andrei Shleifer, and Robert Vishny (1998). "A model of investor sentiment". In: *Journal of Financial Economics* 49.3, pp. 307–343. ISSN: 0304-405X.
- Barrero, Jose Maria (2022). "The micro and macro of managerial beliefs". In: *Journal of Financial Economics* 143.2, pp. 640–667. ISSN: 0304-405X.
- Belo, Frederico, Xiaoji Lin, and Santiago Bzdresch (2014). "Labor Hiring, Investment, and Stock Return Predictability in the Cross Section". In: *Journal of Political Economy* 122.1, pp. 129–177.
- Belo, Frederico et al. (Feb. 2023). "What Drives Firms' Hiring Decisions? An Asset Pricing Perspective". In: *The Review of Financial Studies*. hhad012. ISSN: 0893-9454.
- Bhandari, Anmol, Jaroslav Borovička, and Paul Ho (May 2024). "Survey Data and Subjective Beliefs in Business Cycle Models". In: *The Review of Economic Studies*, rdae054. ISSN: 0034-6527.
- Bianchi, Francesco, Cosmin Ilut, and Hikaru Saijo (Feb. 2023). "Diagnostic Business Cycles". In: *The Review of Economic Studies*, rdad024. ISSN: 0034-6527.

# References II

- Bianchi, Francesco, Sydney C. Ludvigson, and Sai Ma (2022). "Belief Distortions and Macroeconomic Fluctuations". In: *American Economic Review* 112.7, pp. 2269–2315.
- Bianchi, Francesco, Sydney C Ludvigson, and Sai Ma (2024). *What Hundreds of Economic News Events Say About Belief Overreaction in the Stock Market*. Working Paper 32301. National Bureau of Economic Research.
- Bigio, Saki, Dejanir Silva, and Eduardo Zilberman (2025). *Heterogeneous Beliefs, Asset Prices, and Business Cycles*. Working Paper 33672. National Bureau of Economic Research.
- Binsbergen, Jules H van, Xiao Han, and Alejandro Lopez-Lira (Oct. 2022). "Man versus Machine Learning: The Term Structure of Earnings Expectations and Conditional Biases". In: *The Review of Financial Studies* 36.6, pp. 2361–2396. ISSN: 0893-9454.
- Bloom, Nicholas, Mihai A Codreanu, and Robert A Fletcher (2025). *Rationalizing Firm Forecasts*. Working Paper 33384. National Bureau of Economic Research.
- Bordalo, Pedro et al. (Jan. 2021). *Real Credit Cycles*. Working Paper 28416. National Bureau of Economic Research.
- Bordalo, Pedro et al. (2024). "Belief Overreaction and Stock Market Puzzles". In: *Journal of Political Economy* 132.5, pp. 1450–1484.
- Borovičková, Katarína and Jaroslav Borovička (2017). *Discount Rates and Employment Fluctuations*. 2017 Meeting Papers 1428. Society for Economic Dynamics.
- Chaudhry, Aditya (2025). "The impact of prices on analyst cash flow expectations: Reconciling subjective beliefs data with rational discount rate variation". In: *Journal of Financial Economics* 171, p. 104095. ISSN: 0304-405X.
- Chen, Long, Zhi Da, and Xinlei Zhao (Feb. 2013). "What Drives Stock Price Movements?" In: *The Review of Financial Studies* 26.4, pp. 841–876. ISSN: 0893-9454.

# References III

- Chodorow-Reich, Gabriel and Loukas Karabarbounis (2016). "The Cyclicalities of the Opportunity Cost of Employment". In: *Journal of Political Economy* 124.6, pp. 1563–1618.
- Coibion, Olivier and Yuriy Gorodnichenko (2015). "Information rigidity and the expectations formation process: A simple framework and new facts". In: *American Economic Review* 105.8, pp. 2644–78.
- De La O, Ricardo, Xiao Han, and Sean Myers (Jan. 2022). "The Cross-section of Prices and Long-term Returns". In: *SSRN Electronic Journal*.
- De La O, Ricardo and Sean Myers (2021). "Subjective Cash Flow and Discount Rate Expectations". In: *The Journal of Finance* 76.3, pp. 1339–1387.
- Décaire, Paul H and John R Graham (2024). "Valuation fundamentals". In: *Available at SSRN 4951338*.
- Donangelo, Andres (2014). "Labor Mobility: Implications for Asset Pricing". In: *The Journal of Finance* 69.3, pp. 1321–1346.
- Donangelo, Andres et al. (2019). "The cross-section of labor leverage and equity returns". In: *Journal of Financial Economics* 132.2, pp. 497–518. ISSN: 0304-405X.
- Du, William et al. (2025). *Perceived Unemployment Risks over Business Cycles*. Staff Working Papers 2025-23. Bank of Canada.
- Elsby, Michael W. L. and Ryan Michaels (2013). "Marginal Jobs, Heterogeneous Firms, and Unemployment Flows". In: *American Economic Journal: Macroeconomics* 5.1, 1–48.
- Evans, George W. and Seppo Honkapohja (2001). *Learning and Expectations in Macroeconomics*. Princeton University Press. ISBN: 9780691049212.

# References IV

- Favilukis, Jack and Xiaoji Lin (Aug. 2015). "Wage Rigidity: A Quantitative Solution to Several Asset Pricing Puzzles". In: *The Review of Financial Studies* 29.1, pp. 148–192. ISSN: 0893-9454.
- Fukui, Masao, Niels Joachim Gormsen, and Kilian Huber (2024). *Sticky Discount Rates*. Working Paper 32238. National Bureau of Economic Research.
- Gabaix, Xavier (2019). "Behavioral inattention". In: *Handbook of Behavioral Economics - Foundations and Applications* 2. Ed. by B. Douglas Bernheim, Stefano DellaVigna, and David Laibson. Vol. 2. Handbook of Behavioral Economics: Applications and Foundations 1. North-Holland, pp. 261–343.
- Giglio, Stefano et al. (2021). "Five Facts about Beliefs and Portfolios". In: *American Economic Review* 111.5, 1481–1522.
- Greenwood, Robin and Samuel G. Hanson (Nov. 2014). "Waves in Ship Prices and Investment \*". In: *The Quarterly Journal of Economics* 130.1, pp. 55–109. ISSN: 0033-5533.
- Greenwood, Robin and Andrei Shleifer (2014). "Expectations of returns and expected returns". In: *The Review of Financial Studies* 27.3, pp. 714–746.
- Hagedorn, Marcus and Iouri Manovskii (2008). "The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited". In: *American Economic Review* 98.4, 1692–1706.
- Hall, Robert E. (Feb. 2017). "High Discounts and High Unemployment". In: *American Economic Review* 107.2, pp. 305–30.
- Hall, Robert E. and Paul R. Milgrom (2008). "The Limited Influence of Unemployment on the Wage Bargain". In: *American Economic Review* 98.4, 1653–74.
- Jin, Lawrence J. and Pengfei Sui (2022). "Asset pricing with return extrapolation". In: *Journal of Financial Economics* 145.2, Part A, pp. 273–295. ISSN: 0304-405X.

# References V

- Kehoe, Patrick J., Virgiliu Midrigan, and Elena Pastorino (2019). “Debt Constraints and Employment”. In: *Journal of Political Economy* 127.4, pp. 1926–1991.
- Kehoe, Patrick J et al. (Aug. 2023). “Asset Prices and Unemployment Fluctuations: A Resolution of the Unemployment Volatility Puzzle”. In: *The Review of Economic Studies* 90.3, pp. 1304–1357. ISSN: 0034-6527.
- Kilic, Mete and Jessica A Wachter (Jan. 2018). “Risk, Unemployment, and the Stock Market: A Rare-Event-Based Explanation of Labor Market Volatility”. In: *The Review of Financial Studies* 31.12, pp. 4762–4814. ISSN: 0893-9454.
- Kudlyak, Marianna (2014). “The cyclicalities of the user cost of labor”. In: *Journal of Monetary Economics* 68, pp. 53–67. ISSN: 0304-3932.
- Kuehn, Lars-Alexander, Mikhail Simutin, and Jessie Jiaxu Wang (2017). “A Labor Capital Asset Pricing Model”. In: *The Journal of Finance* 72.5, pp. 2131–2178.
- Liu, Yukun (2021). “Labor-based asset pricing”. In: *SSRN*.
- Ljungqvist, Lars and Thomas J. Sargent (2017). “The Fundamental Surplus”. In: *American Economic Review* 107.9, 2630–65.
- Ma, Yueran et al. (2020). *A Quantitative Analysis of Distortions in Managerial Forecasts*. Working Paper 26830. National Bureau of Economic Research.
- Malmendier, Ulrike and Stefan Nagel (Oct. 2015). “Learning from Inflation Experiences”. In: *The Quarterly Journal of Economics* 131.1, pp. 53–87. ISSN: 0033-5533.
- Mankiw, N. Gregory and Ricardo Reis (Nov. 2002). “Sticky Information versus Sticky Prices: A Proposal to Replace the New Keynesian Phillips Curve”. In: *The Quarterly Journal of Economics* 117.4, pp. 1295–1328. ISSN: 0033-5533.

# References VI

- Marcet, Albert and Thomas J Sargent (1989). "Convergence of least squares learning mechanisms in self-referential linear stochastic models". In: *Journal of Economic Theory* 48.2, pp. 337–368. ISSN: 0022-0531.
- Meeuwis, Maarten et al. (2023). *Time-Varying Risk Premia, Labor Market Dynamics, and Income Risk*. Working Paper 31968. National Bureau of Economic Research.
- Menzio, Guido (2023). "Stubborn Beliefs in Search Equilibrium". In: *NBER Macroeconomics Annual* 37, pp. 239–297.
- Merz, Monika and Eran Yashiv (2007). "Labor and the Market Value of the Firm". In: *American Economic Review* 97.4, 1419–1431.
- Mitra, Indrajit and Yu Xu (Oct. 2019). "Time-Varying Risk Premium and Unemployment Risk across Age Groups". In: *The Review of Financial Studies* 33.8, pp. 3624–3673. ISSN: 0893-9454.
- Nagel, Stefan and Zhengyang Xu (Aug. 2021). "Asset Pricing with Fading Memory". In: *The Review of Financial Studies* 35.5, pp. 2190–2245. ISSN: 0893-9454.
- (Feb. 2022). *Dynamics of Subjective Risk Premia*. Working Paper 29803. National Bureau of Economic Research.
- Petrosky-Nadeau, Nicolas, Lu Zhang, and Lars-Alexander Kuehn (Aug. 2018). "Endogenous Disasters". In: *American Economic Review* 108.8, pp. 2212–45.
- Pissarides, Christopher A. (2009). "The Unemployment Volatility Puzzle: Is Wage Stickiness the Answer?" In: *Econometrica* 77.5, pp. 1339–1369.
- Shimer, Robert (Mar. 2005). "The Cyclical Behavior of Equilibrium Unemployment and Vacancies". In: *American Economic Review* 95.1, pp. 25–49.

# References VII

- Sims, Christopher A. (2003). "Implications of rational inattention". In: *Journal of Monetary Economics* 50.3. Swiss National Bank/Study Center Gerzensee Conference on Monetary Policy under Incomplete Information, pp. 665–690. ISSN: 0304-3932.
- Timmermann, Allan G. (1993). "How Learning in Financial Markets Generates Excess Volatility and Predictability in Stock Prices". In: *The Quarterly Journal of Economics* 108.4, pp. 1135–1145. ISSN: 00335533, 15314650.
- Venkateswaran, Venky (2014). "Heterogeneous information and labor market fluctuations". In: *Available at SSRN* 2687561.
- Woodford, Michael (2001). *Imperfect Common Knowledge and the Effects of Monetary Policy*. Working Paper 8673. National Bureau of Economic Research.